Final batch (CC-F): what we tested and found

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What this batch was about

We asked one question: can a model's "self" (what it says it will do, or says it values) control what it does in situations it hasn't seen? The short answer from 18 experiments: what steers the model is the wording in view when it acts. Framing that wording as "I" adds little.

- **When:** the last batch of the character-and-conduct programme in the MATS exploration sprint, from the night of 28 October to the evening of 29 October 2026. Friday is for write-up only.
- **Budget:** about $202 of real money across RunPod, OpenRouter and Modal.
- **Main bet A, training:** teach Qwen3-14B to write a forecast of its own behaviour and then act, and see whether that self-forecast keeps it honest on new agent tasks. A later version trained a value sentence instead: "As an assistant, I must not…"
- **Main bet B, pressure:** if role, authority or persona pressure pushes a model to misbehave, can capping or self-recognition stop it? This needed a pressure effect first.
- **In-context tests (A0):** the same questions without training, by putting a statement in the prompt. We varied who wrote it, "I" against "Assistants", and an accurate description of the model's own habit.
- **Evening checks:** robustness, replications on new scenarios and other models, and a causal test of the one "self" lead.

The test settings, in plain words

Three kinds of task carry every result below. Most numbers are a violation rate, and changes are in percentage points.

Setting	What the model does	What we measure
ODCV agent tasks (ODCV-Bench)	Works in a small fake workplace with files and shell tools, under pressure to hit a target. It can hit the target by breaking a rule, such as editing applicants' resumes or deleting records.	Whether it commits that scenario's named violation ("V1"), read by an automatic file check that agrees with two judge families (κ 0.91). A "scenario" is one task in one of two variants.

Setting	What the model does	What we measure
The desk	Makes one decision on a case: report it, refer it, or impose a penalty itself (the rule says don't).	How often it picks the penalise-it-myself option. One model version has a trained-in habit of doing so (W), with a twin (Z) and the base model.
ImpossibleBench	Fixes code whose tests can only pass by cheating.	Whether it cheats on the tests.

- **Models:**
  - Qwen3-14B for all training and the desk;
  - Qwen3.6-27B for most prompt tests;
  - gpt-oss-20b and MiniMax M2.7 as models from other labs.
- **Introspection adapter:** a released add-on that is meant to make a model describe its own trained-in behaviour. We tested whether it reveals or erases a real self-report.
- **Agent-task sandboxes:** Modal at first, then our own stand-in on RunPod once Modal's credit ran out. The stand-in reproduced Modal's start states and outputs exactly.

Results, experiment by experiment

Changes are in percentage points of violations, with 95% intervals in brackets. A negative number means fewer violations. "Untestable" or "descriptive" means the pre-registered test couldn't give a verdict, so the numbers are a lead, not a finding.

Experiment	Question, in plain words	What happened	Verdict
A0 desk (Qwen3-14B)	Does a forecast in view steer the act? Does it matter who wrote it?	The act follows whatever option the forecast names (96–98% vs 1–3%), whoever wrote it. An accurate habit note ("you choose X about N% of the time") raises the habit by 26–28. Adding "so I'll check the rule first" doesn't fix it. "Do not choose X" cuts it by 39–45.	Done
A0.1 value lines (Qwen3.6-27B, 13 scenarios)	Does one value line cut a violation it never names? Does "I" help?	Every value line cut it, by 12–31 from 56%. The named rule cut 35. "I'm an assistant who…" vs "Assistants…": −4.8 [−11, +1]. "Must not" beat "doesn't" by 13.5 (spotted after the data).	Done
A0G (gpt-oss-20b, 26 scenarios)	The same on another lab's model	Only "As an assistant, I must not…" helped: −11.5, with the interval reaching +1. The other lines did nothing.	Done, model-specific
R1 robustness (existing data)	Is any headline carried by one scenario?	No. Value lines cut violations in 10 of 13 scenarios, with 1 the other way. The habit note raises the habit in 27 of 40 desk cases, with 1 down. Dropping any one scenario keeps every result.	Holds

Experiment	Question, in plain words	What happened	Verdict
R2 (Qwen3.6-27B, 25 new scenarios)	Does the value line hold beyond A0.1, and in the user turn?	System prompt: −30 [−46, −14]. User turn: −32. Where it sits made no difference (−2).	Extends
R3 (Qwen3.6-27B, 13 scenarios)	Does "you tend to [violation]" backfire on agent tasks, as on the desk?	−9 [−26, +7], so no backfire; it leans towards a cut. "Don't [violation]" cut 52.	Unclear, no backfire
R5 (MiniMax M2.7, 23 scenarios)	Does the value line work on another lab's model?	The value lines together: −4 [−20, +12]. Even "Don't [violation]" cut only 15 [−35, +4].	Unclear, small
S3 + R4 adapter (Qwen3-14B)	Does the introspection adapter erase a real self-report?	Without it, the risk-trained model calls itself risk-taking 47 points more often than its safe twin. With it, both models say 0. On 30 questions, the adapter's effect is −47 [−63, −32].	Confirmed (one trained model)
B0 pressure gate (Qwen3-14B)	Does pressure raise misbehaviour enough to test defences?	Authority pressure added 2.5 at most. Role pressure added 10, under the 15 needed.	Failed: Bet B skipped
ImpossibleBench (Qwen3-14B)	Does training reduce cheating on impossible tests?	The base model never cheats (0 of 180).	Nothing to measure

Experiment	Question, in plain words	What happened	Verdict
A1 main bet (training Qwen3-14B)	Does forecast-then-act training make the self govern action?	The comparison versions broke: their length-matching filler looped in agent tasks. Forecast-trained vs untrained: −15.5 [−42, +7]. It never wrote a forecast unprompted.	Untestable
A1 rescue	Trained "As an assistant, I must not…" (V5) vs its "Assistants must not…" twin	V5 −17.9 [−35, +0.6]. Both failed the 90% task-completion bar (85% and 74%).	Untestable, a lean
A1 replication (2 more training runs each)	Does the lean hold?	New runs: −14.4. All four runs: −16.1 [−29.5, −0.9]. V5 was lower in all four pairs. Completion still failed.	Lean holds, descriptive
A1 prefill test (one run pair)	Is V5's lead its own opening sentence?	Twin given "As an assistant, I must not…" to start: −39 [−71, −2]. Twin given "Assistants must not…": −33 [−57, −8]. "I" vs "Assistants": −6. V5 given a neutral opening: −8 [−45, +29].	Mixed

Insights

The batch tells one story: the text in view when the model acts steers what it does, and "self" framing adds little to it.

1. **Wording in view steers the act.** On the desk the model takes whatever option a forecast names, whoever wrote it. On agent tasks, one general value line cuts violations it never mentions.

2. **Firm value lines generalise, strongly in Qwen.** Qwen3.6-27B: −23 to −31 across 38 scenarios, in the system prompt or the user turn. gpt-oss-20b and MiniMax M2.7: small. So this is a Qwen result so far.
3. **Naming the action as forbidden is the strongest wording everywhere.** "Don't [violation]" cut 35–52 on Qwen and 15 on MiniMax, more than any value line.
4. **Describing a habit installs it only when the description names an option to pick.** On the desk, an accurate habit note raised the habit by 26–28. On agent tasks, "you tend to [violation]" read more like a warning (−9).
5. **"Self" framing adds little, wherever we put it:**
   - in the prompt: −2 to −5;
   - trained in: −16, descriptive;
   - written into the model's own first turn: −6.

   What helps is having a principle in view; its grammatical person matters little.
6. **Part of trained V5's advantage doesn't come from its sentence.** Blocking its usual opening didn't raise its violations, so training changed something beyond the words it writes.
7. **Training a principle sentence into an agent can cost competence.** The trained models often stated their principle and then didn't act, and how often varied a lot between training runs (completion 58–92%).
8. **An auditing tool can erase a genuine self-report.** The introspection adapter removed the risk model's honest description of its own risk-taking (−47).
9. **Pick models that misbehave before testing defences.** The 14B model barely bent under pressure and never cheated on impossible tests, so two planned tests had nothing to measure.

Caveats

The clearest limit: every training result failed its competence bar, so nothing about trained self-framing is a confirmed finding.

- **Small samples:** 9–38 scenarios per test and few repeats, so most intervals are 20–40 points wide.
- **Mostly one model family:** most positive results are on Qwen. The two other-lab models showed small effects.
- **Training runs vary a lot:** the same recipe differed by 11–23 points between runs. With 1–4 runs per version, a single pair can mislead.
- **The competence confound:** trained versions finished 58–92% of tasks. Violation rates mix finished runs with looping or stalled ones.

- **Some findings came after the data:** "must not" beating "doesn't", the finished-only reads, and the opening-sentence correlation. These are leads to test, not results.
- **The prefill test lacks one control:** there was no twin-with-a-neutral-opening arm, so "a principle in view" isn't fully separated from "any pre-written opening". The model also may not tell a pre-written turn from its own.
- **One measured violation per scenario:** the file check reads the scenario's named violation only. The judge label for other misconduct failed (κ 0.08), so other rule-breaking isn't counted.
- **The desk isn't an agent task:** it's a one-step choice on models with a trained-in habit. Its backfire didn't carry over to agent tasks.
- **The adapter result rests on one trained risk model,** and its 20 new questions are rewordings of the same trait.
- **Two setup differences:** the RunPod stand-in matched Modal's start states and outputs, but blocks only name lookups, not all network access; no run reached the network. MiniMax's scenarios were picked from ODCV's own judge labels, not from our check.
- **Thinking settings differ:** Qwen3.6-27B ran with thinking on, Qwen3-14B with it off.

What to do next

The most useful follow-up is a clean test of trained self-commitment, because it is the only open path to the ambitious claim.

1. **A competence-preserving training recipe.** For example, mix in agent trajectories, place the principle only at the start of a turn, and train for fewer steps. Use at least three training runs per wording, and add the missing twin-with-a-neutral-opening arm.
2. **The value line's reach beyond Qwen:** larger open models and frontier models with headroom on ODCV, longer runs, and memory files instead of the system prompt.
3. **Habit descriptions in agent memory:** when does a description of past behaviour install the behaviour, and when does it work as a warning? This bears directly on deployed memory systems.
4. **Adapter erasure across traits and models,** to see whether the introspection adapter's erasure is general or specific to the risk trait.
5. **Pressure only where models bend:** find a model whose violations pressure actually raises before testing capping or self-recognition.

Spend and set-up notes

The batch booked $173 of about $202.

Provider	Spent	What it paid for
RunPod	$65	training and serving pods, sandbox hosts
OpenRouter	$63	Qwen3.6-27B, gpt-oss and MiniMax agent runs, judges
Modal	$46	agent-task sandboxes until its account limit was hit at 07:22 on 29 October

- **Left at the end:** about $10 on RunPod and $4.94 on OpenRouter.
- **Modal stand-in:** a proot-based replacement, run on RunPod pods. It was validated on every pod before use: start states identical to Modal's, and released benchmark runs replayed with the same output on every command.
- **Host quirks fixed along the way:**
  - the package installer hangs inside proot unless its sandbox user is root;
  - one pod's older kernel needed proot's seccomp mode turned off.
- **Cost estimates missed twice:** A1's Modal use ran at $32 against a $12 estimate, and R2/R3 at $13.9 against my $4–5. Estimate from earlier full runs, not from a short test.
- **Where things are:** every experiment has its folder, pre-registration tag and RESULT.md under `character_conduct/final_bets/`, and the folder README holds the status table.